12-09 第一篇

Genetics and DNA: The Blueprint of Life

DNA is often called the blueprint of life. It is a molecule found in almost every cell of every living thing. DNA carries the instructions that tell cells how to grow, develop, and function. Your eye colour, height, and many health traits are influenced by the genes stored in your DNA. Understanding DNA helps us understand why we look and behave the way we do. [1]

Genes are sections of DNA that hold specific information. You inherit half your genes from each parent, which is why you may look like your mother or father. Some traits, like blood type, are controlled by a single gene. Others, like height or intelligence, come from many genes working together. The mix of genes you receive is unique to you. [2]

Scientists have learned to read and even edit DNA. This has led to advances in medicine, such as finding genes linked to diseases and developing new treatments. In agriculture, DNA technology helps create crops that resist pests or grow in difficult conditions. Some questions about using this power—such as editing human embryos—remain highly debated. [3]

Genetics also explains similarities and differences between species. Humans share about 98% of their DNA with chimpanzees. Even plants and bacteria have DNA. By comparing DNA across species, scientists can trace how life has evolved over millions of years. This knowledge helps us see our place in the natural world. [4]

DNA research continues to open new doors. It can help solve crimes, identify ancient remains, and improve food production. At the same time, we must use this knowledge responsibly. The more we learn about genetics, the more important it becomes to balance science with ethics. [5]

选项

A. In this way, DNA is at the heart of what makes each living thing unique. B. This inheritance explains both family resemblance and individual variation. C. These advances bring great benefits but also raise important ethical questions. D. Such comparisons reveal how all life on Earth is connected. E. For this reason, genetics will shape both our future and the choices we must make. F. For example, the BRCA gene is linked to a higher risk of certain cancers.

答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

深度解析

- [1] A：
  - **线索**：空前介绍DNA为生命蓝图、存在于几乎所有细胞、携带指令控制生长发育与功能、影响眼色身高与健康性状、理解DNA有助于理解外貌与行为，空后需总结DNA与个体独特性的关系。A的"In this way"承接前文，"at the heart of what makes each living thing unique"与前文对应。
- [2] B：
  - **线索**：空前说基因是DNA片段承载特定信息、从父母各继承一半、单基因控制血型等、多基因共同作用决定身高智力等、每个人的基因组合独特，空后需总结遗传与家族相似及个体差异的关系。B的"This inheritance"指代前文，"family resemblance and individual variation"与前文呼应。
- [3] C：
  - **线索**：空前说科学家能读取和编辑DNA、带来医学进步（疾病基因与新疗法）、农业应用（抗虫耐逆境作物）、人类胚胎编辑等仍有争议，空后需总结科技进步与伦理争议并存。C的"These advances"指代前文，"great benefits but also raise important ethical questions"与前文对应。
- [4] D：
  - **线索**：空前说遗传学解释物种异同、人与黑猩猩DNA约98%相似、动植物细菌均有DNA、比较DNA可追溯进化、帮助认识在自然中的位置，空后需总结生命之间的联结。D的"Such comparisons"指代前文，"how all life on Earth is connected"与前文呼应。
- [5] E：
  - **线索**：空前说DNA研究持续开拓——破案、鉴定古遗骸、改进粮食生产、须负责任使用、越了解越需平衡科学与伦理，空后需总结遗传学对未来的影响与抉择。E的"For this reason"承接前文，"genetics will shape both our future and the choices we must make"与前文对应。
- **干扰项分析**：
  - **F. For example, the BRCA gene is linked to a higher risk of certain cancers**：以For example开头，为具体基因案例（BRCA与癌症风险）。各空均为概括或过渡句；[3]易被误选因涉及疾病基因，但该空需总结科技进步与伦理问题的并存，而非补充单一基因的举例。